

Tactics Sheet

Tactics. Below is a list of all the tactics we have across in the workshop thus far.

<code>intro h</code>	Assume the premise of an implication.
<code>exact h</code>	Submit <code>h</code> as a proof of the current goal.
<code>apply h</code>	Apply the result <code>h</code> .
<code>rw [h]</code>	Rewrite using an equality.
<code>rfl</code>	Prove an equality whose two sides are equal by definition.
<code>use k</code>	Choose <code>k</code> as a witness for an existence goal.
<code>cases h</code>	Split a hypothesis into its possible cases.
<code>obtain ⟨k, hk⟩ := h</code>	Extract a witness and its property from an existence hypothesis.
<code>have h : P := by ...</code>	State and prove an intermediate claim.
<code>by_contra h</code>	Assume the goal is false (call this <code>h</code>), to derive a contradiction.
<code>simp</code>	Simplify the goal using standard rewrite rules.
<code>norm_num</code>	Normalize numbers: simplify and prove numerical facts.
<code>positivity</code>	Prove that an expression is positive, nonnegative or nonzero.
<code>linarith</code>	Combine linear equations and inequalities ($\mathbb{N}$, $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{R}$).
<code>nlinarith</code>	Deduce consequences of polynomial equalities and inequalities.
<code>constructor</code>	Split an $\wedge$ goal or prove both directions of $\leftrightarrow$.
<code>ext x</code>	Prove equality by comparing membership or values at x .
<code>change P</code>	Replace the goal by a definitionally equal expression <code>P</code> .
<code>calc</code>	Write a chain of equalities, giving a proof for each step.
<code>ring</code>	Prove a polynomial identity by expanding and collecting terms.